

Problems 14 Algebra 2

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##+TITLE: Problems 14 Algebra 2
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##+OPTIONS: toc:nil
##+LATEX_HEADER: \usepackage{amsmath,amssymb,mathtools}
##+LATEX_HEADER: \newcommand{\im}{\operatorname{im}}
##+LATEX_HEADER: \newcommand{\Span}{\operatorname{span}}
##+LATEX_HEADER: \newcommand{\Ker}{\operatorname{ker}}
##+LATEX_HEADER: \newcommand{\diag}{\operatorname{diag}}

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* Exercise 5.10

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\[
L=\begin{pmatrix}3&2\\0&1\end{pmatrix}
\begin{pmatrix}p&q\\0&r\end{pmatrix}.
\]

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Let $\mathcal{E}$ be a basis of E .

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\[
\mathcal{E}=\left\{
\begin{pmatrix}1&0\\0&0\end{pmatrix},
\begin{pmatrix}0&1\\0&0\end{pmatrix},
\begin{pmatrix}0&0\\0&1\end{pmatrix}
\right\}.
\]

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\[
[L]_{\mathcal{E}}^{\mathcal{E}}=\left(
\left|\begin{matrix}3&0\\0&0\end{matrix}\right|_{\mathcal{E}},
\left|\begin{matrix}0&3\\0&0\end{matrix}\right|_{\mathcal{E}},
\left|\begin{matrix}0&2\\0&1\end{matrix}\right|_{\mathcal{E}}
\right)
=
\begin{pmatrix}
3&0&0\\
0&3&2\\
0&0&1
\end{pmatrix}.
\]

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\[
\det\left([L]_{\mathcal{E}}^{\mathcal{E}}-\lambda I\right)
=\det\begin{pmatrix}
3-\lambda&0&0\\
0&3-\lambda&2\\
0&0&1-\lambda
\end{pmatrix}
=(3-\lambda)^2(1-\lambda).
\]

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\[
\lambda_1=3,\text{qquad}\lambda_2=1.
\]

```

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\[
[L]_{\mathcal{E}}^{\mathcal{E}}
\begin{pmatrix}p\\q\\r\end{pmatrix}
=
\begin{pmatrix}
3&0&0\\
0&3&2\\
0&0&1
\end{pmatrix}
\begin{pmatrix}p\\q\\r\end{pmatrix}
=
\begin{pmatrix}3p\\3q+2r\\p\end{pmatrix}
=\lambda\begin{pmatrix}p\\q\\r\end{pmatrix}.
\]

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If $\lambda=3$, then $2r=0$, so $r=0$.

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\[
V_3^{\mathcal{E}}=\text{Span}\left\{
\begin{pmatrix}1\\0\\0\end{pmatrix}_{\mathcal{E}},
\begin{pmatrix}0\\1\\0\end{pmatrix}_{\mathcal{E}}
\right\}.
\]

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For $\lambda=1$, $p=0$, $q=-r$.

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\[
V_1^{\mathcal{E}}=\text{Span}\left\{
\begin{pmatrix}0\\1\\-1\end{pmatrix}_{\mathcal{E}}
\right\}.
\]

```

\]

So,

\[

$V_1 \oplus V_3 = E \Rightarrow L$ is diagonalizable.

\]

* Exercise 5.14

Using consequence of Theorem 5.18.

$v_1, \dots, v_n$ are eigenvectors assigned to

\[

$\lambda_1 \neq \dots \neq \lambda_n$.

\]

They are linearly independent.

So they form a basis of E .

\[

$\Rightarrow f$ is diagonalizable. $\square$

\]

* Exercise 5.16

Since P is projection,

\[

$P^2 = P$,

\]

and

\[

$S^2 = I$.

\]

1. Let λ be an eigenvalue of P .

Then

\[

$Px = \lambda x$

\]

for some $x \neq 0$.

Then

\[

$P^2x = Px = \lambda x = P(\lambda x) = \lambda Px = \lambda^2 x$.

\]

So

\[

$\lambda^2 = \lambda$,

\]

so

\[

$\lambda \in \{0, 1\}$.

\]

1. Let μ be an eigenvalue of S .

So

\[

$Sy = \mu y$

\]

for some $y \neq 0$.

Then

\[

$S^2y = Iy = y = \mu^2 y$.

\]

So

\[

$\mu^2 = 1 \Rightarrow \mu \in \{-1, 1\}$.

\]

1. Since P is projection operator, it is defined with respect to some subspace, precisely $\text{Im } P$.

So if $\lambda = 0$ is eigenvalue of P ,

\[

$Px = 0$

\]

for some x , hence $x \in \text{Ker } P$; in such case

\[

$V_0^P = \text{Ker } P$.

\]

If $\lambda = 1$ is eigenvalue,
 $\backslash$
 $P y = y$
 $\backslash$
for some y .

But we know that P works like identity on subspace with respect to which P has been defined.

So
 $\backslash$
 $V_1^P = \text{im } P$.
 $\backslash$

By rank-nullity,
 $\backslash$
 $V_1^P \oplus V_0^P = \text{Ker } P \oplus \text{im } P = E$.
 $\backslash$

Hence P is diagonalizable.

Now let's talk about S .

Suppose S is defined with respect to some subspace U , s.t.
 $\backslash$
 $E = U \oplus F \rightarrow \forall x \in E, \quad x = x_U + x_F$,
 $\backslash$
 $\backslash$
 $S(x) = x_U - x_F$.
 $\backslash$

Then for $\lambda = 1$,
 $\backslash$
 $Sx = x$
 $\backslash$
for some x , hence
 $\backslash$
 $x \in U \rightarrow V_1^S = U$.
 $\backslash$

For $\lambda = -1$,
 $\backslash$
 $Sy = -y$
 $\backslash$
for some y , then
 $\backslash$
 $y \in F \rightarrow V_{-1}^S = F$.
 $\backslash$

So
 $\backslash$
 $V_1^S \oplus V_{-1}^S = U \oplus F = E \rightarrow S$ is diagonalizable.
 $\backslash$

* Exercise 5.17

1.
 $\backslash$
 $A = \begin{pmatrix} x_n & \dots & x_{n-1} \end{pmatrix}$
 $=$
 $\begin{pmatrix} x_{n+1} & \dots & x_n \end{pmatrix}$
 $=$
 $\begin{pmatrix} x_n & \dots & x_{n-1} \end{pmatrix}$.
 $\backslash$

So
 $\backslash$
 $A = \begin{pmatrix} 1 & \dots & 1 \end{pmatrix}$.
 $\backslash$

1.
 $\backslash$
 $\det(A - \lambda I) = (1 - \lambda)(-\lambda) - 1 = \lambda^2 - \lambda - 1 = 0$.
 $\backslash$

$\backslash$
 $\lambda_1 = \frac{1 + \sqrt{5}}{2}, \quad \lambda_2 = \frac{1 - \sqrt{5}}{2}$.
 $\backslash$

Those eigenvalues are distinct, A is diagonalizable.

Let $\varphi = \lambda_1$ and $\psi = \lambda_2$.

$$D = \begin{pmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{pmatrix}.$$

Let's find eigenbasis now.

$$\begin{pmatrix} 1 & 1 \\ x & y \end{pmatrix} = \lambda \begin{pmatrix} x & y \end{pmatrix}.$$

So

$$x = \lambda y$$

and

$$x + y = \lambda x.$$

Take $y=1$, then $x=\lambda$ and

$$\lambda + 1 = \lambda^2,$$

which is true for eigenvalues.

So eigenspace assigned to λ is

$$\text{Span}\left\{\begin{pmatrix} \lambda \\ 1 \end{pmatrix}\right\}.$$

So

$$V_{\lambda_1}^A = \text{Span}\left\{\begin{pmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{pmatrix}\right\},$$

$$V_{\lambda_2}^A = \text{Span}\left\{\begin{pmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{pmatrix}\right\}.$$

Then

$$P = \begin{pmatrix} \frac{1+\sqrt{5}}{2} & \frac{1-\sqrt{5}}{2} \\ 1 & 1 \end{pmatrix}$$

is eigenbasis.

So

$$P^{-1} = \frac{1}{\det P} \operatorname{adj}(P).$$

The adjugate is C^T , where C is cofactor matrix.

$$C = \begin{pmatrix} 1 & -1 \\ -\psi & \varphi \end{pmatrix}$$

$$\rightarrow \operatorname{adj}(P) = \begin{pmatrix} 1 & -\psi \\ -1 & \varphi \end{pmatrix}.$$

Also

$$\det P = \sqrt{5}.$$

So

$$P^{-1} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -\psi \end{pmatrix}$$

-1φ
 $\end{pmatrix}.$
 $\backslash]$

Then
 $\backslash[$
 $A=PDP^{-1}.$
 $\backslash]$

Hence
 $\backslash[$
 $A^n\begin{pmatrix}x_1\\x_0\end{pmatrix}$
 $=$
 $(PD^nP^{-1})\begin{pmatrix}x_1\\x_0\end{pmatrix}.$
 $\backslash]$

$\backslash[$
 $=\frac{1}{\sqrt{5}}$
 $\begin{pmatrix}\varphi&\psi\\1&\psi\end{pmatrix}$
 $\begin{pmatrix}\varphi^n&0\\0&\psi^n\end{pmatrix}$
 $\begin{pmatrix}1&-\psi\\1&\varphi\end{pmatrix}$
 $\begin{pmatrix}x_1\\x_0\end{pmatrix}$
 $\backslash]$

$\backslash[$
 $=\frac{1}{\sqrt{5}}$
 $\begin{pmatrix}\varphi^{n+1}&\psi^{n+1}\\\varphi^n&\psi^n\end{pmatrix}$
 $\begin{pmatrix}1&-\psi\\1&\varphi\end{pmatrix}$
 $\begin{pmatrix}x_1\\x_0\end{pmatrix}$
 $\backslash]$

$\backslash[$
 $=\frac{1}{\sqrt{5}}$
 $\begin{pmatrix}\varphi^{n+1}-\psi^{n+1} & -\psi\varphi^{n+1}+\varphi\psi^{n+1}\\\varphi^n-\psi^n & -\psi\varphi^n+\varphi\psi^n\end{pmatrix}$
 $\begin{pmatrix}x_1\\x_0\end{pmatrix}$
 $\backslash]$

$\backslash[$
 $=\frac{1}{\sqrt{5}}$
 $\begin{pmatrix}\varphi^{n+1}-\psi^{n+1}+x_0(-\psi\varphi^{n+1}+\varphi\psi^{n+1})\\x_1(\varphi^n-\psi^n)+x_0(-\psi\varphi^n+\varphi\psi^n)\end{pmatrix}$
 $\end{pmatrix}.$
 $\backslash]$

So,
 $\backslash[$
 $x_n=\frac{x_1(\varphi^n-\psi^n)+x_0(-\psi\varphi^n+\varphi\psi^n)}{\sqrt{5}}.$
 $\backslash]$